



```
from collections import deque
import time
def water_jug_bfs(jug1,jug2,target):
    start_time=time.time()
    visited=set()
    queue=deque()
    queue.append((0,0))
    while queue:
        a,b=queue.popleft()
        if (a,b) in visited:
            continue
        visited.add((a,b))
        if a == target or b == target:
            print("solution found:",(a,b))
            print("step_Taken:",len(visited))
            print("time taken:",time.time()-start_time)
            return
        queue.extend([
            (jug1,b),(a,jug2),(0,b),(a,0),
            (a-min(a,jug2-b),b+min(a,jug2-b)),
            (a+min(b,jug1-a),b-min(b,jug1-a))
        ])

water_jug_bfs(4,3,2)
```

```
... solution found: (4, 2)
step_Taken: 10
time taken: 0.00011849403381347656
```